Do **not** write solutions on this page.

12. [Maximum mark: 20]

Consider the family of functions f_n defined by $f_n(x)=\sum_{r=0}^{n}\left(-2x^2\right)^r$, where $x \in \mathbb{R}$ and $n \in \mathbb{N}$.

(a) Show that f_n is an even function for all values of n. [3]

(b) (i) Show that $f_3(x) = 1 - 2x^2 + 4x^4 - 8x^6$.

(ii) Write down a similar expression for $f_4(x)$ in ascending powers of x. [2]

Consider the function $f(x)=\lim_{n\to\infty} f_n(x)$ defined over the domain $-k < x < k$ where $k > 0$.

The largest possible value of k is K.

(c) (i) Find the value of K, giving your answer in exact form.

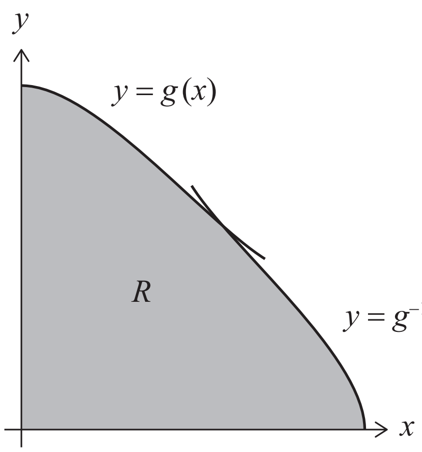
(ii) Express $f(x)$ as a rational function in the form $\frac{1}{a+bx^2}$, where a and b are constants to be determined. [5]

The function g is defined as $g(x)=f(x)$ for $0 \le x < K$.

(d) (i) Justify that g^{-1} exists.

(ii) Find $g^{-1}(x)$, giving its domain. [6]

The region R is completely enclosed by the curves $y=g(x)$, $y=g^{-1}(x)$ and the x- and y-axes, as shown on the following diagram.



(e) Find the area of R. [4]